



PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 05 – Discriminative Classifiers II: Perceptrons and Decision Trees



Chizhi Chris ZHANG

zhangchizhi@ciomp.ac.cn

Advanced Computing and Digital Technology Research Center

University of Chinese Academy of Sciences

Fall 2026

A user marks one urgent looking message as spam.

Move one linear boundary because the current prediction was wrong.

Ask which condition should split similar messages next.

```
graph LR; Evidence[EVIDENCE  
Training mistakes] --> Representation[REPRESENTATION  
Weights or splits]; Representation --> Method[METHOD  
Classifier update]; Method --> Decision[DECISION  
Flag or release]; Decision --> Evidence;
```

Outcome and error feed the next observation

How do mistakes update a rule & questions partition a space?

Syllabus boundary for Lecture 05

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Syllabus topic	Allocated time	Class emphasis
Perceptron algorithm	1.0 hour	updates, convergence, failure
Decision trees	1.5 hours	partition, split, pruning
Integrated practice	0.5 hour	coding and case reasoning



Deliberate boundary

Feature engineering begins in Lecture 06. Random forests wait until the ensemble lectures.

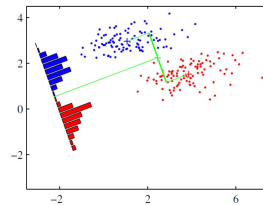
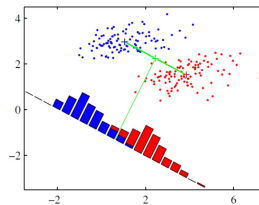
Two boundary languages

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Model	Prediction language	Typical geometry
Perceptron	$\text{sign}(w^T x + b)$	one hyperplane
Decision tree	a path of feature tests	axis aligned regions

Same data

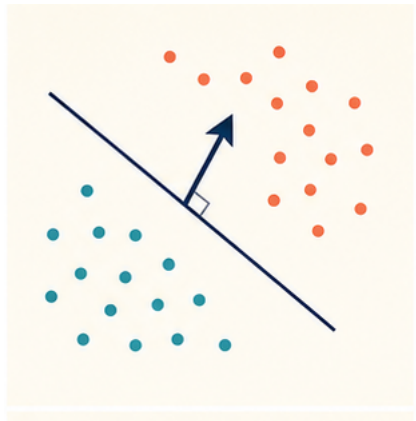
The two models may fit different patterns and fail for different reasons.



$$a(x) = w^\top x + b$$

Predict class +1 when $a(x) > 0$, otherwise predict class -1.

w points across the boundary and b shifts it.





$$\tilde{X} = [X^\top, \mathbf{1}]^\top$$

Augmented weight

$$\tilde{w} = [w^\top, b]^\top$$

$$a(x) = \tilde{w}^\top \tilde{x}$$

Use $y_i \in \{-1, +1\}$.

$$y_i \tilde{w}^\top \tilde{x}_i > 0$$

A correct prediction makes the signed score positive.

$y_i \tilde{w}^\top \tilde{x}_i > 0$ gives no change.

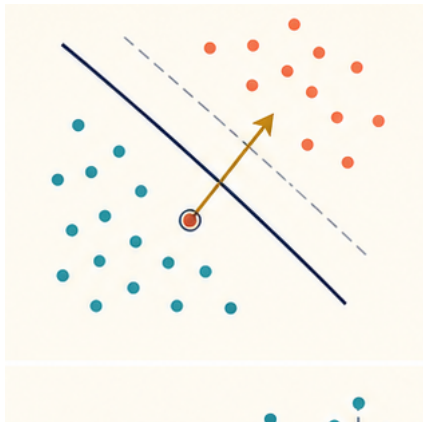
$$y_i \tilde{w}^\top \tilde{x}_i \leq 0$$

$$\tilde{W} \leftarrow \tilde{W} + \eta y_i \tilde{X}_i$$

A positive message lies on the negative side.

Add ηx to w and add η to b .

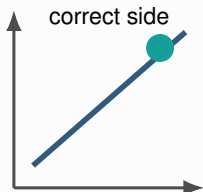
One sample changes the score of every future sample.



Reward and punishment as an early metaphor

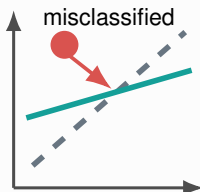
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Correct sample



no update

Mistake



move the boundary

Historical wording

Correct patterns receive no penalty. Misclassified patterns push the weight vector.

Modern reading

The rule optimizes a classification criterion. It does not imitate a biological neuron in detail.

A four point training set

工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Sample	$\tilde{x} = (x_1, x_2, 1)$	y
A	(0,0,1)	+1
B	(0,1,1)	+1
C	(1,0,1)	-1
D	(1,1,1)	-1



Initial state

$\tilde{w}^{(0)} = (0, 0, 0)$ and $\eta = 1$.

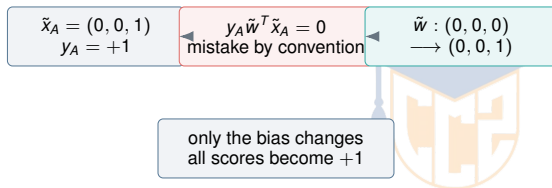
The first mistake creates a bias

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

$$y_A \tilde{w}^{(0)\top} \tilde{x}_A = 0$$
$$\tilde{w}^{(1)} = (0, 0, 0) + (0, 0, 1) = (0, 0, 1)$$

Meaning

The classifier now predicts every point as positive.



A negative mistake changes the slope

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

$$y_c \tilde{w}^T \tilde{x}_c = (-1)(1) = -1$$
$$\tilde{w} \leftarrow (0, 0, 1) - (1, 0, 1) = (-1, 0, 0)$$

**Boundary**

$$-x_1 = 0$$

Several epochs can be necessary

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Epoch

One pass through all training samples.

Stopping test

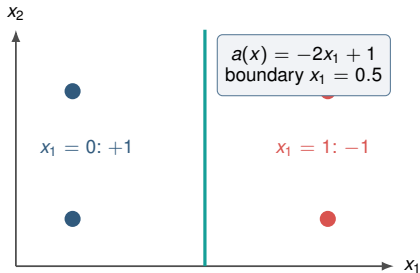
Finish an entire epoch with zero mistakes.

Bad shortcut

Stopping after a lucky local run can leave earlier samples misclassified.

The learned boundary in the classroom example

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER



The second feature receives zero weight because it is irrelevant in this four-point example.

One valid solution

$$\tilde{w} = (-2, 0, 1)$$

$$a(x) = -2x_1 + 1$$



Check

$x_1 = 0$ gives $+1$ and $x_1 = 1$ gives -1 .

Perceptron learning loop

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

```
import numpy as np
X = np.array([[0.,0.,1.],[0.,1.,1.],[1.,0.,1.],[1.,1.,1.]])
y = np.array([1., 1., -1., -1.])
w = np.zeros(3)

for epoch in range(20):
    mistakes = 0
    for xi, yi in zip(X, y):
        if yi * (w @ xi) <= 0:
            w += yi * xi
            mistakes += 1
    print(epoch, mistakes, w)
    if mistakes == 0:
        break
```

Expected pattern

Mistakes decrease to zero for this separable data.

Learning rate changes the path

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Update size

η controls how far one mistake moves the weight vector.

Separable case

A fixed positive learning rate preserves the convergence guarantee.

Finite budget

With early stopping, different learning rates can leave different classifiers.

Sample order changes the solution

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Same constraints

Every training point demands a positive signed score.

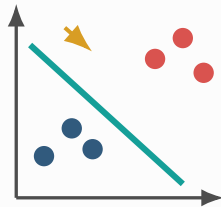
Different path

The next violated constraint depends on sample order.

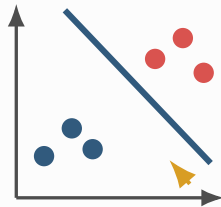
Consequence

Two valid runs can end at different separating hyperplanes.

Order A



Order B



Convergence needs linear separability

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Guarantee

For linearly separable data, repeated mistake updates reach a separating weight vector in finitely many mistakes.



Condition

The guarantee disappears when no separating hyperplane exists.

Margin explains an informal mistake bound

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Data radius

Assume $\|x_i\| \leq R$.

Separating margin

Assume a unit vector u with $y_i u^\top x_i \geq \gamma > 0$.

$$M \leq (R/\gamma)^2$$

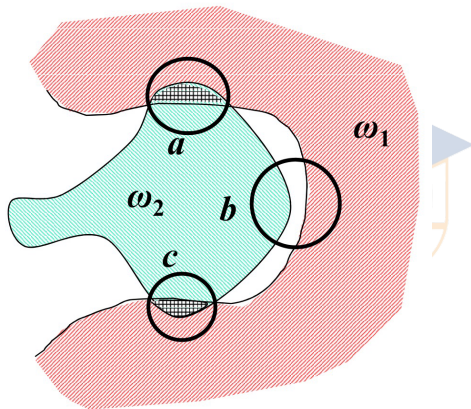
Exclusive OR makes the perceptron cycle

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

x_1	x_2	y
0	0	-1
0	1	+1
1	0	+1
1	1	-1

Geometry

No straight line separates the two positive corners from the two negative corners.



Cycling is a diagnostic signal

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Observed pattern

Training mistakes repeat without reaching zero.

- 1 Check label errors and duplicate rows with conflicting labels.
- 2 Plot low dimensional projections.
- 3 Try a richer representation or a different model.



Do not claim convergence

A small recent mistake count is not a separating solution.

The pocket rule keeps the best visit

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Idea

During a nonconvergent run, remember the weight vector with the fewest training mistakes.

Benefit

The final epoch no longer automatically decides the returned classifier.

Limit

Training error still needs validation on unseen data.

Averaging reduces path noise

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Averaged perceptron

Return the average of visited weight vectors rather than only the last vector.

Intuition

Short lived oscillations cancel while persistent directions remain.

Still linear

Averaging does not solve a genuinely nonlinear boundary.



Multiclass perceptron uses one score per class

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Scores

$$a_k(x) = w_k^\top x$$

$$\hat{y} = \arg \max_k a_k(x)$$



Interpretation

Each class owns a direction that competes for the sample.

A multiclass mistake updates two rows

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

If $\hat{y} \neq y$

$$w_y \leftarrow w_y + \eta X$$

Competing class

$$w_{\hat{y}} \leftarrow w_{\hat{y}} - \eta X$$

Effect

Raise the true class score and lower the mistaken winner.

One versus rest is a different construction

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Construction	Training	Possible issue
Multiclass perceptron	joint competing scores	relative margins
One versus rest	independent binary models	scores may not align



Do not merge the names

Both use linear scores, but their training objectives differ.

Feature scale can dominate updates

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Two features

Age ranges from 18 to 70 while annual spending ranges from 0 to 100000.

Update

The change ηyx inherits the units and scale of every feature.

Consequence

A large numeric range can dominate the path without being more informative.

Bias must follow the same update convention

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Separate form

$$w \leftarrow w + \eta yx \text{ and } b \leftarrow b + \eta y$$

Augmented form

$$\tilde{w} \leftarrow \tilde{w} + \eta y \tilde{x}$$

Common bug

Updating w but forgetting b forces the boundary through the origin.

Prediction at exactly zero needs one rule

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Option A

Treat $a(x) = 0$ as a mistake during training.

Option B

Break the prediction tie with a fixed class convention.

Requirement

Use the same convention in hand calculations, code, and evaluation.

```
def train_perceptron(X, y, epochs=30, eta=1.0):
    w = np.zeros(X.shape[1])
    for epoch in range(epochs):
        mistakes = 0
        for xi, yi in zip(X, y):
            if yi * (w @ xi) <= 0:
                w = w + eta * yi * xi
                mistakes += 1
        if mistakes == 0:
            return w, epoch + 1
    return w, epochs
```

Mutation

Replace `<=` with `<` and test a zero initialization.

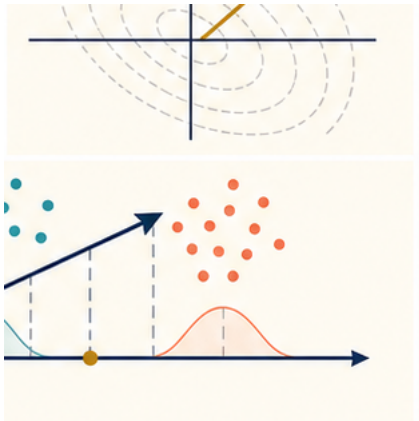
Perceptron lessons that survive the formula

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

- Mistakes drive learning.
- Linear separability supports convergence.
- Order and scale affect the path.
- A zero training error does not guarantee future performance.

Model boundary

A linear learner needs a richer representation when the pattern is nonlinear.



A hospital triage tree

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Decision-tree case: from split to action

First question

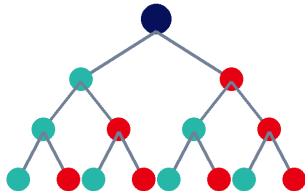
Is oxygen saturation below 92 percent?

Second question

If not, is breathing rate above the age adjusted threshold?

Leaf action

Urgent review, routine observation, or discharge advice.



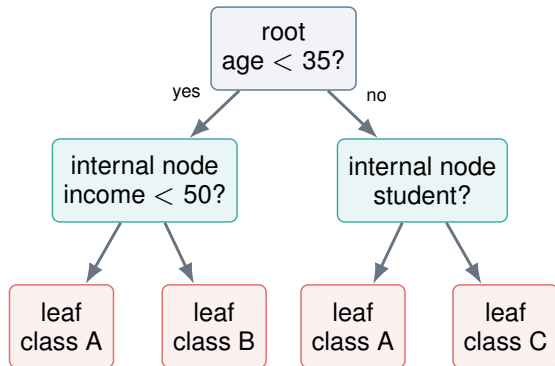
Trace one sample

Report the leaf
distribution

Test nearby
counterfactuals

Root, internal nodes, branches, and leaves

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER



A branch records a test outcome; a leaf returns the prediction.

Root

The first test sees every training sample.

Internal node

A feature condition routes a subset.

Leaf

A prediction summarizes the samples that arrive there.

100

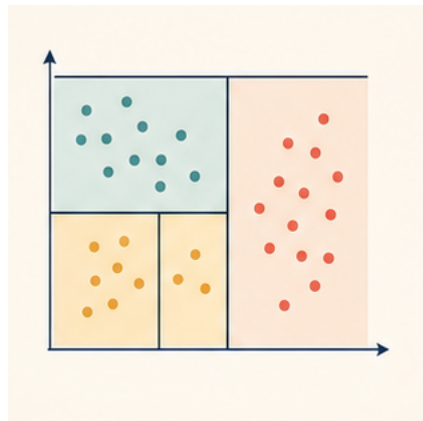
Conditions on different branches do not all apply to one sample.

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

A condition such as $x_i \leq t$ cuts one current region into two parts.

Later tests divide only the samples that reach that node.

In two dimensions, numeric tests create axis aligned rectangles.



$$\hat{p}_{mk} = \frac{n_{mk}}{n_m}$$

$$\hat{y}_m = \arg \max_k \hat{p}_{mk}$$

A pure leaf with two samples gives a fragile probability estimate.

$$\hat{c}_m = \frac{1}{n_m} \sum_{i: x_i \in R_m} y_i$$

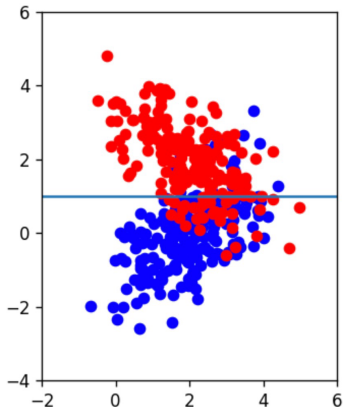
$$\hat{f}(x) = \sum_m \hat{c}_m \mathbb{I}(x \in R_m)$$

The prediction changes in steps when a sample crosses a split.

A split reads like a simple threshold rule.

A diagonal boundary may require many small rectangles.

Construct a rotated feature or use a model with oblique boundaries.



Payment method has card, cash, transfer, and voucher values.

Send a subset of categories left and the remaining categories right.

Invalid shortcut

Integer codes do not create a meaningful order for nominal categories.

Impute before fitting the tree.

Use a model that learns a default branch for missing values.

Missingness may itself carry information and may change after deployment.

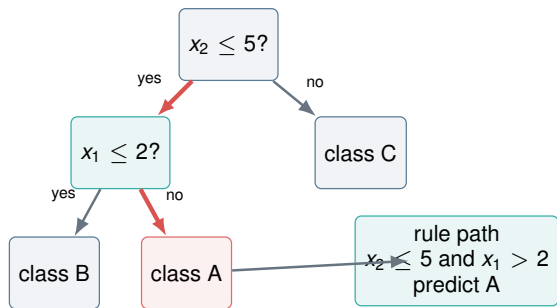
The same binary questions can separate several classes step by step.

Each leaf may contain counts for every class.

A fault diagnosis tree can end in normal, sensor fault, motor fault, or overload.

A borrowed tree example becomes a rule list

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER



Read the diagram

Start at the top test, follow one branch result, and stop at a leaf label.

Translation

If $x_2 \leq 5$ and then $x_1 > 2$, predict the leaf class shown on that path.

Overclaim

The path does not prove that changing debt ratio will cause the outcome to change in reality.

Tree rules translate directly into code

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

```
def manual_tree(oxygen, breathing_rate):  
    if oxygen < 92:  
        return "urgent review"  
    if breathing_rate > 24:  
        return "observe"  
    return "routine advice"
```

Trace

oxygen=95 and breathing_rate=27 reaches the second return.

Depth controls rule complexity

Depth	Typical behavior	Risk
1	one stump	underfitting
3 to 5	short interactions	often readable
very deep	small specialized leaves	high variance

No universal best depth

Validation data must judge the tradeoff.

Deep trees have low bias and high variance

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Low bias

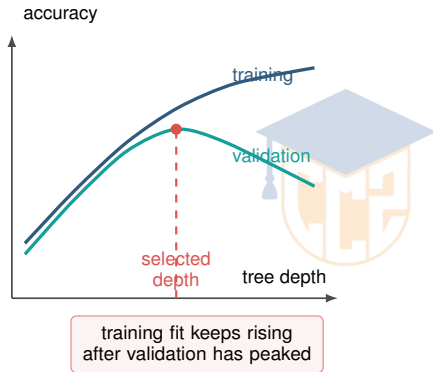
A sufficiently deep tree can fit intricate training patterns.

High variance

Small data changes can alter early splits and the entire subtree below them.

Symptom

Training accuracy stays high while validation accuracy falls.



The best root split is oxygen < 92.

After one borderline case changes, breathing rate becomes the best root split.

1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 2680, 26

A path can look precise even when the tree structure is unstable.

Stopping rules limit fragmentation

- Maximum depth limits path length.
- Minimum samples for a split limits tiny parent nodes.
- Minimum samples per leaf limits fragile terminal groups.
- Minimum impurity decrease rejects weak splits.

Tune together

Different rules can produce similar sized trees for different reasons.

Minimum leaf size protects probability estimates

Leaf counts	Estimated positive rate	Reliability
2 of 2	1.00	very uncertain
40 of 50	0.80	more stable
400 of 500	0.80	still population dependent

Same purity, different evidence

Count the observations behind every leaf probability.

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Only 2 percent of transactions are fraudulent.

Give costly or rare classes more influence when evaluating splits.

Meaning

Weights encode an objective choice. They do not create new positive examples.

A shallow tree in Python

```
from sklearn.tree import DecisionTreeClassifier

tree = DecisionTreeClassifier(
    criterion="gini",
    max_depth=3,
    min_samples_leaf=20,
    random_state=7,
)
tree.fit(X_train, y_train)
print(tree.get_depth(), tree.get_n_leaves())
print(tree.score(X_train, y_train))
print(tree.score(X_valid, y_valid))
```

Read together

Depth, leaf count, training score, and validation score describe different parts of the fit.

Text rules expose the fitted structure

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

```
from sklearn.tree import export_text

rules = export_text(tree, feature_names=list(feature_names))
print(rules)
```

Inspection question

Do the root feature and thresholds make sense in the measurement units?

Tree tracing checkpoint

Applicant

income=8, debt ratio=0.62, prior defaults=0

- 1 Follow the root test.
- 2 Write every condition on the selected path.
- 3 Report the leaf sample count and class ratio.
- 4 Name one nearby change that reaches a different leaf.



Complete answer

A class label without a path and leaf evidence is incomplete.

A useful split makes children more homogeneous

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Parent node

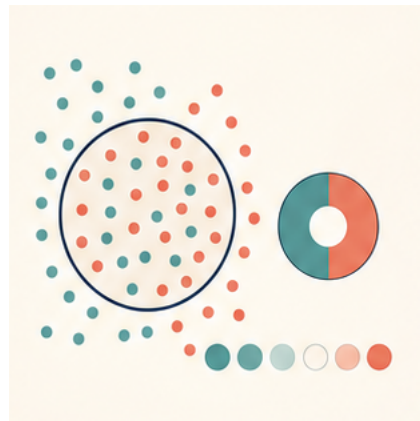
Several classes are mixed together.

Candidate split

A feature threshold sends samples into two children.

Goal

The weighted child impurity should be lower than the parent impurity.



Entropy measures label uncertainty

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

$$H(R) = - \sum_{k=1}^K p_k \log_2 p_k$$

Pure node

$H = 0$ when one class has probability 1.

Balanced binary node

$H = 1$ bit when $p_1 = p_2 = 0.5$.

Gini impurity is a squared probability measure

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

$$G(R) = 1 - \sum_{k=1}^K p_k^2$$

Binary form

$$G = 2p(1 - p).$$

Range

A pure node gives 0. A balanced binary node gives 0.5.

Useful for describing leaf error.

Split growth

It may treat distinct probability mixtures as equally impure.

Three impurities on one binary node

Positive share	Entropy	Gini	Error
0.50	1.000	0.500	0.500
0.80	0.722	0.320	0.200
1.00	0.000	0.000	0.000

Shared ordering

All three rank these nodes from most mixed to pure.

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

The split removes much label uncertainty.

Child sizes belong in the formula.

Gini decrease follows the same weighting pattern

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

$$\Delta G = G(R) - \frac{n_L}{n} G(R_L) - \frac{n_R}{n} G(R_R)$$



Tree choice

Select the feature and threshold with the largest decrease among current candidates.

A hand calculation for one split

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Node	Positive	Negative	Gini
Parent	4	4	0.500
Left	3	1	0.375
Right	1	3	0.375

$$\Delta G = 0.5 - \frac{4}{8}(0.375) - \frac{4}{8}(0.375) = 0.125$$



A perfect tiny child may give modest gain

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Child	Size	Positive share
Left	1	1.00
Right	99	0.49



Visual trap

The pure child looks impressive but represents only one observation.

Decision

Use the weighted impurity decrease and minimum leaf constraints.

Consider midpoints between adjacent distinct sorted values.

For 2, 3, 7, and 10, candidates include 2.5, 5, and 8.5.

Do not split identical values into different branches.

Sorted scans reuse cumulative counts

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Naive search

Recount every class on both sides for every threshold.

Efficient scan

Move one sorted sample at a time and update left and right class counts.

Result

Every threshold can be scored in one pass after sorting.

A feature with K categories has many possible binary partitions.

Exhaustive subset search becomes expensive when cardinality is high.

Group rare categories, use ordered statistics carefully, or choose a model with native categorical handling.

A classification tree commonly minimizes Gini impurity.

A regression tree commonly minimizes squared error within children.

Every internal node has two outgoing branches.

Regression splits reduce within node squared error

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

$$E(R) = \sum_{i: x_i \in R} (y_i - \bar{y}_R)^2$$

$$\Delta E = E(R) - E(R_L) - E(R_R)$$



Leaf prediction

Each child predicts its own mean.

Evaluate candidates and keep the largest immediate impurity reduction.

A standard greedy tree does not revisit an early split after growing later nodes.

100

Fast construction can miss a better tree whose first step looks weaker.

The globally best tree is computationally hard

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Search space

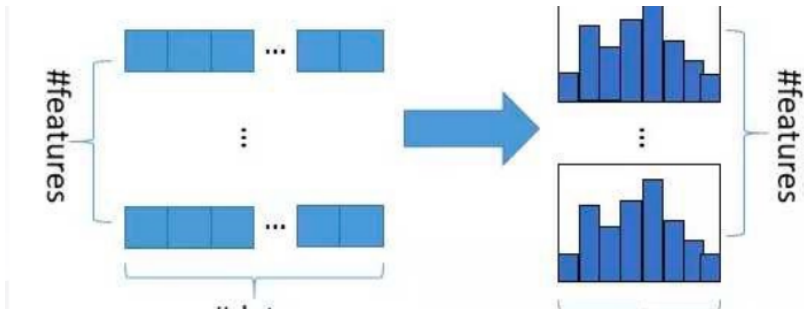
Tree structures, features, thresholds, and stopping points interact.

Practical strategy

Grow greedily, constrain complexity, and validate or prune.

Claim

The fitted tree is a useful heuristic solution, not proof of a globally optimal structure.



Search. Accumulate class or gradient statistics per bin, then test bin boundaries.

Benefit. Large datasets need fewer candidate thresholds.

A split search in plain Python

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

```
def gini(y):
    if len(y) == 0:
        return 0.0
    p = np.bincount(y, minlength=2) / len(y)
    return 1.0 - np.sum(p ** 2)

def split_gain(x, y, threshold):
    left = x <= threshold
    n = len(y)
    child = left.sum()/n * gini(y[left])
    child += (~left).sum()/n * gini(y[~left])
    return gini(y) - child
```

Unit test

A perfect separation should give a larger gain than a random threshold.

Exclusive OR is easy for a small tree

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

First split

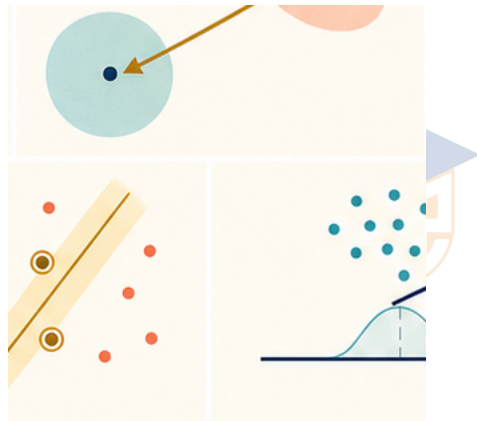
Test $x_1 \leq 0.5$.

Second level

Within each side, test $x_2 \leq 0.5$.

Result

Four leaves represent the four corners exactly.



A fully grown tree can memorize noise

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Training data

A mislabeled customer becomes a single pure leaf.

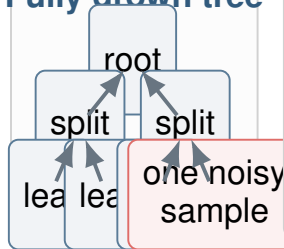
Future data

The special rule has no stable support and fails on similar customers.

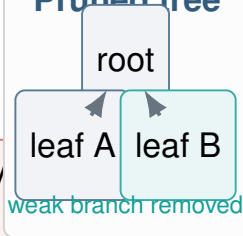
Response

Remove weak branches or prevent them from growing.

Fully grown tree



Pruned tree



ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Depth, leaf size, split size, and minimum impurity decrease.

Training remains faster and the tree stays compact.

An early weak split may enable valuable later structure, so stopping can be premature.

Grow a relatively rich tree.

Evaluate nested subtrees and remove branches whose benefit does not justify their complexity.

Advantage

Later evidence can judge whether an early split became useful.

Cost complexity pruning balances fit and size

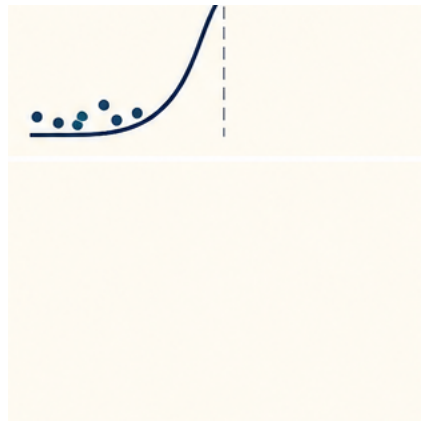
$$R_{\alpha}(T) = R(T) + \alpha |T|_{\text{leaves}}$$

Fit term

$R(T)$ measures training error or impurity.

Complexity term

α charges every leaf.



The pruning path contains nested subtrees

Small alpha

Keep more branches.

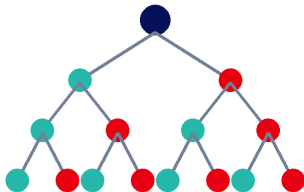
Larger alpha

Collapse the weakest link and obtain a smaller subtree.

Path

Candidate trees are nested rather than unrelated random structures.

Pruning & honest tree evaluation



Pre-pruning

Post-pruning

Validation selects complexity

Validation selects the subtree

- 1 Fit candidate pruning paths on training folds.
- 2 Evaluate each candidate on held out folds.
- 3 Choose the complexity rule.
- 4 Refit on the available training data.
- 5 Evaluate once on the untouched test set.

Leakage

Pruning on the test set turns the test set into training data.

Training and validation curves tell different stories

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Tree size	Training error	Validation error
small	high	high
moderate	lower	lowest
very large	near zero	higher



Selection

The validation minimum identifies useful complexity, not the training minimum.

Class frequency within a leaf.

Deep leaves produce extreme 0 or 1 estimates from few samples.

100

Use calibration curves and proper scoring rules on validation data.

Laplace smoothing tempers tiny leaves

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

$$\tilde{p}_k = \frac{n_k + \alpha}{n + K\alpha}$$

Binary leaf

For 2 positives out of 2 and $\alpha = 1$, the estimate becomes 3/4 rather than 1.

Limit

Smoothing helps small counts but does not fix dataset shift.

Add impurity decreases from every split that uses a feature.

Continuous or high cardinality features offer more candidate splits and may appear important by chance.

Check

Compare with validation permutation importance and domain knowledge.

Tree explanations need nearby counterexamples

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Path explanation

List the conditions that led to the leaf.

Counterexample

Change one feasible feature and trace whether the route changes.

Feasibility

A mathematically small change may be impossible in the real system.

Bootstrap trees reveal structural stability

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Procedure

Refit the same tree specification on several bootstrap samples.

Inspect

Track root features, common paths, and validation performance.

Finding

Similar scores with different roots signal predictive usefulness but weak structural certainty.

CASE EVIDENCE TO ACTION

**OBSERVE**

Training mistakes

**USE**

Pruning & honest tree evaluation

FAILURE TO TEST

Pruning on the final test set.

RESOLVE

Why can a smaller tree generalize better?

Perceptron and tree assumptions

Question	Perceptron	Decision tree
Boundary	one hyperplane	rectangular regions
Learning signal	misclassified samples	impurity decrease
Main risk	nonseparability	high variance
Typical control	representation and stopping	depth and pruning

Choice

Match the model language to the pattern and the cost of mistakes.

A global weighted score summarizes risk.

A short sequence of conditions separates operational groups.

Use the same train, validation, and test partitions for a fair contest.

```
graph TD; 1[1 Augmented vectors & weight space] --> 2[2 The perceptron algorithm]; 2 --> 3[3 Recursive tree partitions]; 3 --> 4[4 Split criteria & pruning]; 4 --> 5[5 Pruning & honest tree evaluation]; 5 --> 6[6 Decision-tree case: from split to action];
```

- 1 Augmented vectors & weight space
- 2 The perceptron algorithm
- 3 Recursive tree partitions
- 4 Split criteria & pruning
- 5 Pruning & honest tree evaluation
- 6 Decision-tree case: from split to action

Coding comparison on the same split

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

```
from sklearn.linear_model import Perceptron
from sklearn.tree import DecisionTreeClassifier

models = {
    "perceptron": Perceptron(max_iter=1000, random_state=7),
    "tree": DecisionTreeClassifier(max_depth=3, min_samples_leaf=10,
                                   random_state=7),
}

for name, model in models.items():
    model.fit(X_train, y_train)
    print(name, model.score(X_train, y_train),
          model.score(X_valid, y_valid))
```

Next step

Inspect errors and structure after the score comparison.

Integrated case for an inspection gate

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Data

Temperature, vibration, operating mode, and 300 labeled inspections.

- 1 Build a scaled perceptron baseline.
- 2 Build a shallow pruned tree.
- 3 Compare costly missed faults.
- 4 Inspect stability after resampling.



Deployment

Record how missing sensors route through the system.

Failure diagnosis by symptom

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Symptom	Likely cause	First check
Perceptron cycles	not separable or bad labels	plot and audit labels
Tree trains perfectly	excess complexity	validation gap and leaf sizes
Tree root keeps changing	correlated features or small data	bootstrap stability
High score disappears	leakage or shift	pipeline and time split

Perceptron and tree self-check

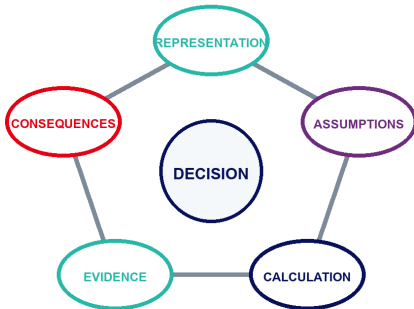
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

- 1 Explain one perceptron update in words.
- 2 State the condition behind perceptron convergence.
- 3 Compute a weighted Gini decrease.
- 4 Translate one root to leaf path into a rule.
- 5 Explain why pruning must not use the test set.

Strong answer

Use a concrete sample, one equation, and one failure condition.

A CHECKABLE DECISION LOOP



How do mistakes update a rule & questions partition a space?

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

We learned how two classifiers use a feature vector.

We will decide how raw values become valid, useful, and leakage free features.

Bridge

The same raw timestamp can become an hour, a weekday, a cyclic pair, or a source of leakage.